

QSECDEF

Quantum Security and Defence

PROFESSIONAL CERTIFICATION PROGRAMME

Certified Quantum Business Professional

CQBP — Understand Quantum. Understand the Risk. Be Ready to Lead.

Quantum is moving from emerging technology to business risk — and organisations will need people who understand what comes next.

Post-quantum cryptography is entering enterprise technology roadmaps. Governments and regulators are increasing their focus on quantum readiness. New capabilities are emerging across computing, communications, sensing and security.

For business, cybersecurity, risk and technology professionals, understanding quantum is rapidly becoming a practical professional requirement.

The Certified Quantum Business Professional (CQBP) programme from Quantum Security and Defence (QSECDEF) is designed to give professionals a structured understanding of the quantum landscape — without requiring a background in quantum physics.

10

WEEKS

20

HOURS

10

SESSIONS

CQBP

CERTIFICATION

THE PROGRAMME

10 weeks. 20 hours. One complete quantum business foundation.

The programme consists of 10 weekly two-hour sessions, taking participants from the fundamentals of quantum technology through to its business, cybersecurity, regulatory and strategic implications.

The emphasis is not simply on understanding what quantum is. It is about understanding what quantum changes — and what organisations need to do about it.

WHAT YOU WILL LEARN

01 Introduction to the Quantum Era

Understand the emerging quantum landscape, the major areas of quantum technology and why quantum is becoming relevant to business, cybersecurity and national security.

02 Quantum Computing Fundamentals

Understand qubits, superposition, entanglement, interference and measurement, together with the principal hardware architectures being developed to build quantum computers.

03 Quantum Algorithms & Applications

Explore how quantum algorithms work and where quantum approaches may create value across optimisation, finance, logistics, chemistry, materials, machine learning and other industries.

04 Post-Quantum Cryptography

Understand the quantum threat to existing public-key cryptography, Harvest Now Decrypt Later, emerging PQC standards and the challenge of migrating complex organisations toward quantum-safe cryptography.

05 Quantum Key Distribution & Quantum Networks

Explore QKD, secure quantum key exchange, quantum communications, network architectures and the developing technologies behind future quantum networks.

06 QRNG & Quantum-Safe Cybersecurity

Understand quantum random number generation, entropy and the role of PQC, QKD, QRNG, Zero Trust, defence in depth and crypto agility within future cybersecurity architectures.

07 Quantum Sensing

Discover how quantum technologies can transform navigation, timing, magnetic and gravitational sensing, defence, infrastructure and other precision-measurement applications.

08 Geopolitics, Policy & Regulation

Understand technological sovereignty, export controls, international standards, supply-chain dependencies and the developing regulatory environment surrounding quantum technologies.

09 The Quantum Market & Ecosystem

Explore the emerging commercial quantum ecosystem, investment landscape, technology supply chains, infrastructure requirements, business opportunities and strategic dependencies.

10 PQC Risk & Migration Management

Learn how to move from awareness to action: identifying cryptographic assets, mapping dependencies, assessing vulnerabilities, prioritising risk and developing an executable enterprise PQC migration strategy.

WHO IT IS FOR

Built for professionals — not physicists

CQBP is designed for professionals who need to understand, evaluate, govern or make decisions about quantum technology, rather than build a quantum computer. It is particularly relevant for:



Cybersecurity & CISOs

Understand the cryptographic threat, PQC migration, HNDL and the future of quantum-safe security.



Risk, GRC & Second Line

Understand how quantum creates technology, third-party, concentration, operational resilience and governance risks.



Executives & Board Members

Understand the strategic questions organisations increasingly need to address.



Technology & Architecture Teams

Understand how quantum technologies interact with existing enterprise infrastructure.



Compliance, Legal & Procurement

Understand emerging standards, regulation, supplier dependencies and procurement implications.



Consultants & Professional Services

Build the knowledge required to confidently engage clients on the emerging quantum landscape.

LEARNING OUTCOME

From technology to business decision

By completing the programme, participants will develop an integrated understanding of:



You will leave with the vocabulary and conceptual understanding needed to engage meaningfully with technical specialists, vendors, cybersecurity teams, risk functions and senior leadership.

CERTIFICATION

Earn the CQBP certification

Participants who complete the programme and successfully pass the final examination can earn the:

Certified Quantum Business Professional — CQBP

Awarded by Quantum Security and Defence (QSECDEF).

The certification demonstrates a structured understanding of the technology, cybersecurity, risk, commercial and strategic dimensions of the emerging quantum economy.

THE CASE FOR ACTING

Why now?

The biggest mistake may be waiting for a cryptographically relevant quantum computer before developing quantum capability.

Sensitive information can be collected today for future decryption. Cryptographic migration can take years. Supply-chain and procurement decisions being made today can create dependencies that remain for a decade or more.

And as quantum moves into cybersecurity, regulation, procurement and enterprise strategy, organisations will increasingly need professionals capable of asking the right questions.

The advantage belongs to those who understand the change before they are required to respond to it.

Get ahead of quantum.

10 WEEKS | 20 HOURS | 10 SPECIALIST SESSIONS | PROFESSIONAL CERTIFICATION

Become a Certified Quantum Business Professional.

Join the QSECDEF CQBP programme and build the knowledge to understand what quantum changes — before you are expected to make decisions about it.

Enquire and register: info@qsecdef.com · qsecdef.com